# F = ma in practice

**LEARNING TARGET · R7** I can solve with F = ma — always finding the NET force first.

## SECTION A KEY VOCABULARY

| | |
|---|---|
| **Net force** | Combine opposite forces: subtract and keep the direction of the larger one. |
| **F = ma** | Uses the *net* force, never just one pull. |
| **Direction** | Acceleration points the same way as the net force. |
| **Balanced** | Net force 0 N → a = 0 m/s² (still or constant velocity). |

## SECTION B KEY EQUATION + UNITS

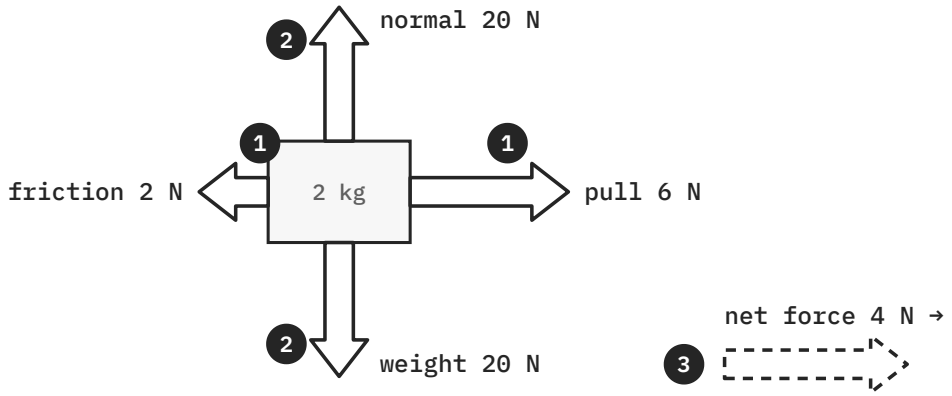
$$\Sigma F_x = m\ a_x$$

$$\Sigma F_x = F_{right} - F_{left}$$

| | |
|---|---|
| F | newtons (N) |
| m · a | kg · m/s² (1 N = 1 kg·m/s²) |

## SECTION C COLOR IT · THE ARROWS COMPETE

*Vertical forces balance.*




☐ **1 · ORANGE** Horizontal forces

☐ **2 · YELLOW** Vertical forces (balanced)

☐ **3 · ORANGE** Net force (thick arrow)

## SECTION D WORKED EXAMPLE

A 4 kg cart has 12 N pulling right and 4 N of friction left. Find the acceleration.

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| m = 4 kg<br>12 N right<br>4 N left | $a = F_{net} \div m$ | $F_{net} = 12 - 4 = 8$ N right<br>$a = 8 \div 4$ | **2 m/s² right**<br>with the net force |

## SECTION E THE 2026-XJ THREAD

DAY 13 OF 22 · NEW TOOL: F = MA

F = ma links it all: from XJ's mass and the net force on it, we can calculate its acceleration, and from that, how its velocity will change.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else